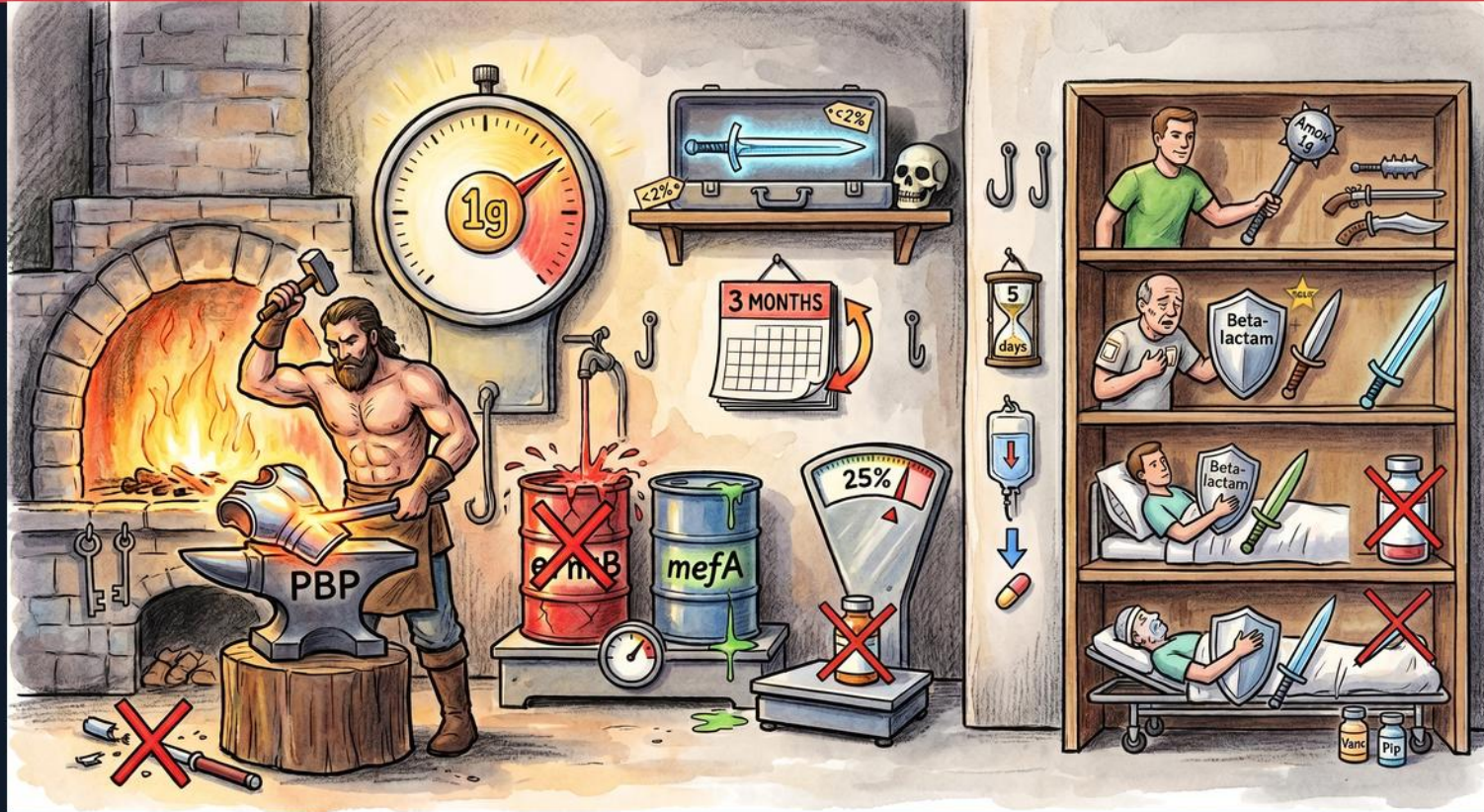


Pneumonia — Resistance & Empiric Antibiotics





PBP anvil (blacksmith)

WHAT YOU SEE

Blacksmith reshaping metal on the 'PBP' anvil — pneumococcus remodels its penicillin-binding protein target.

WHAT IT ENCODES

S. pneumoniae penicillin resistance is altered penicillin-binding proteins (PBPs) — the transpeptidases that build the cell wall. Mosaic *pbp* genes acquired by transformation lower beta-lactam affinity, so more drug is needed. This is TARGET modification, fundamentally different from enzymatic (beta-lactamase) drug destruction.



Broken hammer / wrong tool

WHAT YOU SEE

Discarded tool with a red X on the forge floor — a beta-lactamase inhibitor is the wrong fix for DRSP.

WHAT IT ENCODES

Because pneumococcal resistance is PBP-affinity-based (not enzymatic), adding a beta-lactamase inhibitor like clavulanate does NOTHING for drug-resistant *S. pneumoniae*. The correct fix is to RAISE the dose. Non-meningeal breakpoints were raised (parenteral susceptible ≤ 2 ug/mL) to reflect dose-surmountable resistance.

THE BOARD TRAP

"add clavulanate to beat resistant pneumococcus" -> wrong; resistance is PBP target change, not an enzyme — raise the dose instead.



1g high-dose gauge

WHAT YOU SEE

Big gauge reading '1g' glowing at the dial — high-dose amoxicillin overcomes the low-affinity PBP.

WHAT IT ENCODES

Because resistance is dose-surmountable, most non-meningeal penicillin-'resistant' pneumococci are still treatable with high-dose amoxicillin 1 g PO TID or high-dose IV penicillin/ceftriaxone. Meningeal breakpoints stay strict (≤ 0.06) due to poor CNS penetration.

THE BOARD TRAP

"non-meningeal breakpoint applies to meningitis" -> wrong; use the strict MENINGEAL breakpoint for CNS disease — empiric pneumococcal meningitis is vancomycin + 3rd-gen cephalosporin, not amoxicillin alone.



ermB barrel (X'd)

WHAT YOU SEE

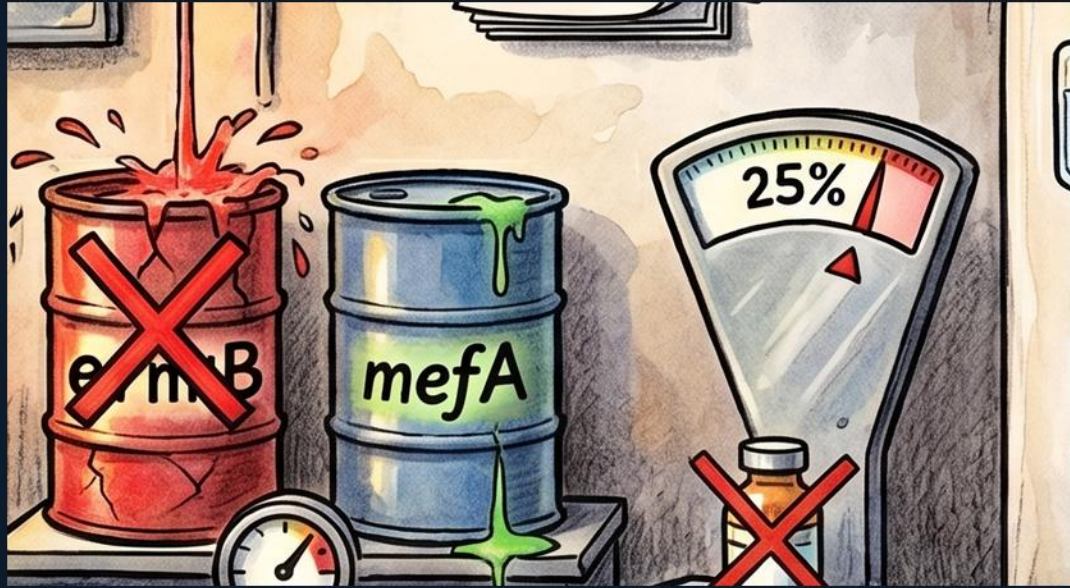
Red barrel labeled 'ermB' with a red X — high-level macrolide resistance by ribosomal methylation.

WHAT IT ENCODES

ermB methylates 23S rRNA (A2058) of the 50S subunit — the macrolide target — giving HIGH-level resistance (MIC ≥ 64) and the MLSB phenotype (cross-resistance to macrolides, lincosamides, streptogramin B). Accounts for ~40-60% of resistant pneumococci in some series.

THE BOARD TRAP

"raise the macrolide dose" for ermB -> wrong; high-level target-site resistance is not dose-surmountable — switch class (beta-lactam, doxycycline, respiratory FQ).



mefA barrel (efflux)

WHAT YOU SEE

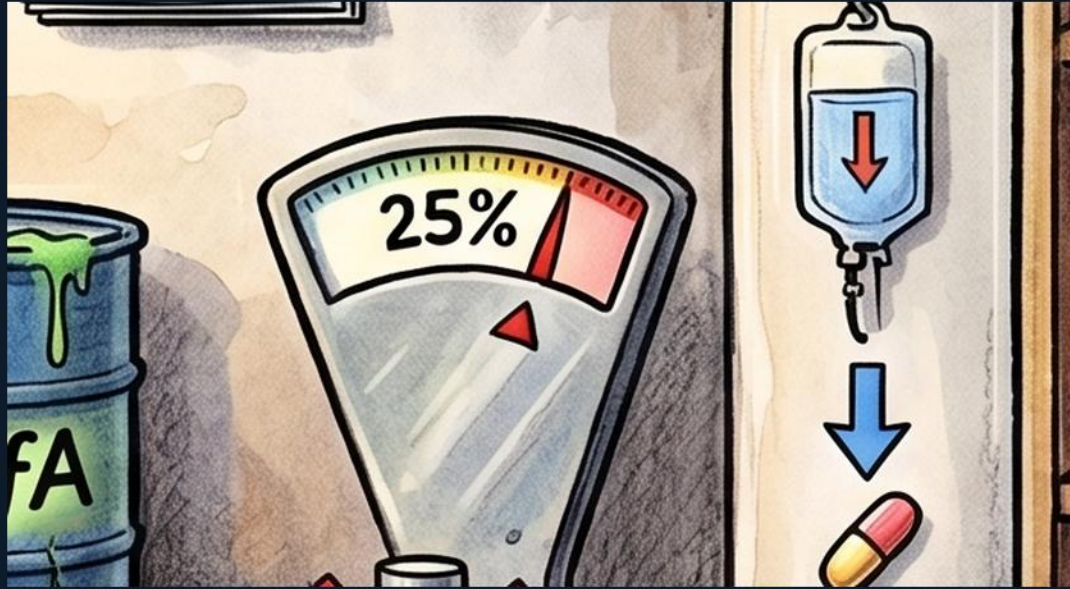
Green barrel labeled 'mefA' with arrows — an efflux pump giving low-level macrolide resistance.

WHAT IT ENCODES

mef(A/E) is a membrane efflux pump that exports 14- and 15-membered macrolides — LOW-level resistance (MIC <16, the M phenotype). Unlike ermB it does NOT affect clindamycin. It is the more common macrolide-resistance mechanism in some regions.

THE BOARD TRAP

"low MIC mef strain is safe for azithromycin monotherapy" -> wrong; even low-level efflux resistance causes clinical failures — use combination therapy or a non-macrolide where resistance is prevalent.



25% resistance scale (X'd)

WHAT YOU SEE

Scale reading '25%' with a red X — the threshold above which macrolide monotherapy is forbidden.

WHAT IT ENCODES

When local pneumococcal macrolide resistance exceeds 25% (US rates ~30-40%), do NOT use a macrolide as monotherapy for CAP. A macrolide is acceptable only as the atypical partner of a beta-lactam, or use a respiratory FQ instead. The 25% threshold is the key testable number.

THE BOARD TRAP

"azithromycin monotherapy for CAP" in the US -> wrong; resistance exceeds 25% almost everywhere — add a beta-lactam or use a respiratory FQ. Monotherapy is OK only for a healthy outpatient with documented low local resistance.



<2% FQ sword on shelf

WHAT YOU SEE

Blue sword tagged '<2%' on the shelf by a skull — respiratory fluoroquinolones, resistance still rare.

WHAT IT ENCODES

Pneumococcal resistance to respiratory FQs (levofloxacin, moxifloxacin) stays low (<2%), so they are reliable single-agent empiric options covering both DRSP and atypicals. FQs inhibit DNA gyrase (gyrA, topo II) and topoisomerase IV (parC); resistance is stepwise gyrA/parC point mutations.

THE BOARD TRAP

"empiric respiratory FQ when TB is plausible" -> wrong (the skull warns); an FQ partially treats and MASKS TB, selecting FQ-resistant TB and delaying diagnosis — avoid with upper-lobe cavitory disease/risk factors. Ciprofloxacin is NOT a respiratory FQ.



3 MONTHS calendar

WHAT YOU SEE

Calendar stamped '3 MONTHS' with a rotation arrow — prior antibiotics within 90 days select resistance.

WHAT IT ENCODES

Antibiotic exposure within the prior 90 days (3 months) is among the strongest risk factors for drug-resistant *S. pneumoniae* and resistant gram-negatives, and an MDR risk factor for HAP/VAP. The recent drug pressures flora toward resistance to THAT class. This single factor often upgrades to a combination/respiratory-FQ regimen.

THE BOARD TRAP

"re-use the recently given class" -> wrong; rotate to a DIFFERENT class — macrolide recently -> beta-lactam/doxy or FQ; beta-lactam recently -> macrolide or FQ.



5-day hourglass / IV drip

WHAT YOU SEE

Hourglass marked '5 days' over an IV drip — minimum duration and the IV-to-PO step-down clock.

WHAT IT ENCODES

Uncomplicated CAP needs a MINIMUM 5 days AND clinical stability (afebrile 48-72 h, stable vitals/oxygenation, eating) before stopping. Longer only for complications: 7+ days MRSA/Pseudomonas; weeks for abscess/empyema/meningitis. Switch IV->PO once hemodynamically stable, improving, able to ingest, with normal GI absorption (FQs, doxy, linezolid, metronidazole are ~100% oral bioavailable).

THE BOARD TRAP

"longer course is better" for stabilized uncomplicated CAP -> wrong; prolonging risks C. difficile/resistance — but don't stop at 5 days if still unstable or complicated.



Top shelf — mace (healthy outpatient)

WHAT YOU SEE

Healthy man swinging a mace on the top shelf — uncomplicated outpatient gets high-dose amoxicillin.

WHAT IT ENCODES

Healthy outpatient, NO comorbidities, no resistance risk: high-dose amoxicillin 1 g PO TID preferred (high dose beats PBP resistance). Doxycycline is an alternative; a macrolide is acceptable only where local resistance <25%. Low mortality; no mandatory atypical combination.

THE BOARD TRAP

"macrolide monotherapy" in a high-resistance region -> wrong even for this lowest-risk group; and don't over-treat a healthy outpatient with a respiratory FQ.



2nd shelf — beta-lactam shield (comorbid)

WHAT YOU SEE

Seated man holding a 'beta-lactam' shield and sword — comorbid outpatient needs beta-lactam plus atypical cover.

WHAT IT ENCODES

Outpatient WITH comorbidities (heart/lung/liver/renal disease, diabetes, alcoholism, malignancy, asplenia) or recent antibiotics: beta-lactam (amox-clavulanate or cephalosporin) PLUS a macrolide or doxycycline, OR a respiratory FQ as monotherapy. Comorbidity raises DRSP risk, so atypical coverage becomes mandatory.

THE BOARD TRAP

"plain amoxicillin monotherapy" with comorbidities -> wrong; no atypical coverage and higher DRSP risk — use beta-lactam + macrolide/doxy or a respiratory FQ alone.



3rd/4th shelf — patients in beds (inpatient)

WHAT YOU SEE

Patients lying in beds on the lower shelves clutching beta-lactam shields — admitted CAP, ward and ICU.

WHAT IT ENCODES

Inpatient non-severe (ward) without MRSA/Pseudomonas risk: beta-lactam (ceftriaxone/cefotaxime/amp-sulbactam) PLUS a macrolide, OR a respiratory FQ alone. Severe/ICU CAP (≥ 1 major or ≥ 3 minor IDSA/ATS criteria): beta-lactam PLUS a macrolide, OR beta-lactam PLUS a respiratory FQ — always a beta-lactam-containing COMBINATION.

THE BOARD TRAP

"respiratory FQ monotherapy for severe/ICU CAP" -> wrong; severe disease needs a beta-lactam-containing combination — FQ alone is inadequate.



X'd bottles — no reflexive broad coverage

WHAT YOU SEE

Pill bottles with red X's at the foot of the beds — don't reflexively add MRSA/Pseudomonas coverage.

WHAT IT ENCODES

Add empiric anti-MRSA (vancomycin/linezolid) or antipseudomonal beta-lactam **ONLY** with specific risk factors (prior MRSA/Pseudomonas isolation, recent IV antibiotics + local validation), then de-escalate by cultures. The abandoned HCAP category drove overtreatment without better outcomes.

THE BOARD TRAP

"treat as HCAP with vanc + pip-tazo" for a stable patient with healthcare contact -> wrong; risk-stratify individually rather than reflexively broadening.

Summary — every symbol

1. PBP anvil (blacksmith)

S

3. 1g high-dose gauge

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5. mefA barrel (efflux)

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8. 3 MONTHS calendar

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Healthy outpatient, NO comorbidities, no resistance risk: high-dose amoxicillin 1 g PO TID preferred (high dose beats PBP resistance)

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